

Finding Area by the Limit Definition

The next theorem shows that the equivalence of the limits (as $n \rightarrow \infty$) of the upper and lower sums is not mere coincidence. It is true for all functions that are continuous and nonnegative on the closed interval $[a, b]$.

THEOREM 4.3 Limits of the Lower and Upper Sums

Let f be continuous and nonnegative on the interval $[a, b]$. The limits as $n \rightarrow \infty$ of both the lower and upper sums exist and are equal to each other. That is,

$$\begin{aligned}\lim_{n \rightarrow \infty} s(n) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(m_i) \Delta x \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(M_i) \Delta x \\ &= \lim_{n \rightarrow \infty} S(n)\end{aligned}$$

where $\Delta x = (b - a)/n$ and $f(m_i)$ and $f(M_i)$ are the minimum and maximum values of f on the i th subinterval.

Finding Area by the Limit Definition

This means that you are free to choose an *arbitrary* x -value in the i th subinterval, as shown in the *definition of the area of a region in the plane*.

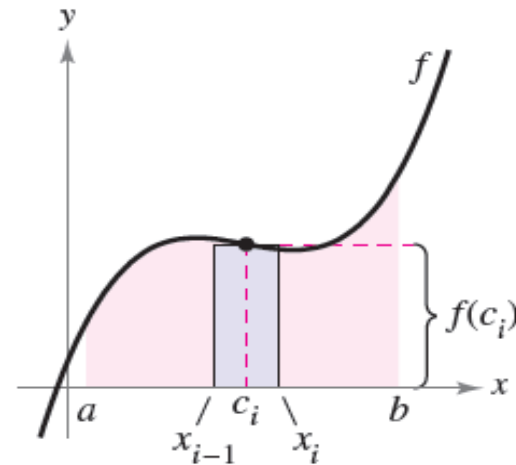
Definition of the Area of a Region in the Plane

Let f be continuous and nonnegative on the interval $[a, b]$. (See Figure 4.13.) The area of the region bounded by the graph of f , the x -axis, and the vertical lines $x = a$ and $x = b$ is

$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x$$

where $x_{i-1} \leq c_i \leq x_i$ and

$$\Delta x = \frac{b - a}{n}.$$

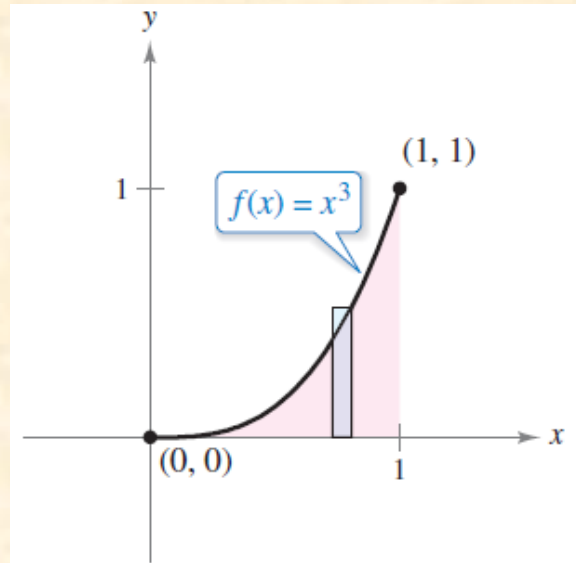


The width of the i th subinterval is $\Delta x = x_i - x_{i-1}$.

Figure 4.13

Example 5 – Finding Area by the Limit Definition

Find the area of the region bounded by the graph of $f(x) = x^3$, the x -axis, and the vertical lines $x = 0$ and $x = 1$, as shown in Figure 4.14.



The area of the region bounded by the graph of f , the x -axis, $x = 0$, and $x = 1$ is $\frac{1}{4}$.

Figure 4.14

Example 5 – Solution

Begin by noting that f is continuous and nonnegative on the interval $[0, 1]$. Next, partition the interval $[0, 1]$ into n subintervals, each of width $\Delta x = 1/n$.

According to the definition of area, you can choose any x -value in the i th subinterval.

For this example, the right endpoints $c_i = i/n$ are convenient (can be any point in the subinterval).

$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{i}{n}\right)^3 \left(\frac{1}{n}\right)$$

Right endpoints: $c_i = \frac{i}{n}$

$$c_i = \text{midpoint of } \left[\frac{i-1}{n}, \frac{i}{n}\right] = \frac{2i-1}{2n}$$

Example 5 – Solution

$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{i}{n}\right)^3 \left(\frac{1}{n}\right)$$

Right endpoints: $c_i = \frac{i}{n}$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^4} \sum_{i=1}^n i^3$$

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^4} \left[\frac{n^2(n+1)^2}{4} \right]$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{4} + \frac{1}{2n} + \frac{1}{4n^2} \right) = \frac{1}{4}$$

The area of the region is $\frac{1}{4}$

Finding Area by the Limit Definition

In general, a good value to choose is the midpoint of the interval, $c_i = (x_{i-1} + x_i) / 2$, and apply the **Midpoint Rule**.

$$\text{Area} \approx \sum_{i=1}^n f\left(\frac{x_{i-1} + x_i}{2}\right) \Delta x. \quad \text{Midpoint Rule}$$

Riemann Sums

Example 1 – A Partition with Subintervals of Unequal Widths

Consider the region bounded by the graph of $f(x) = \sqrt{x}$ and the x-axis for $0 \leq x \leq 1$, as shown in Figure 4.18. Evaluate the limit

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x_i$$

where c_i is the right endpoint of the partition given by $c_i = i^2/n^2$ and Δx_i is the width of the i^{th} interval.

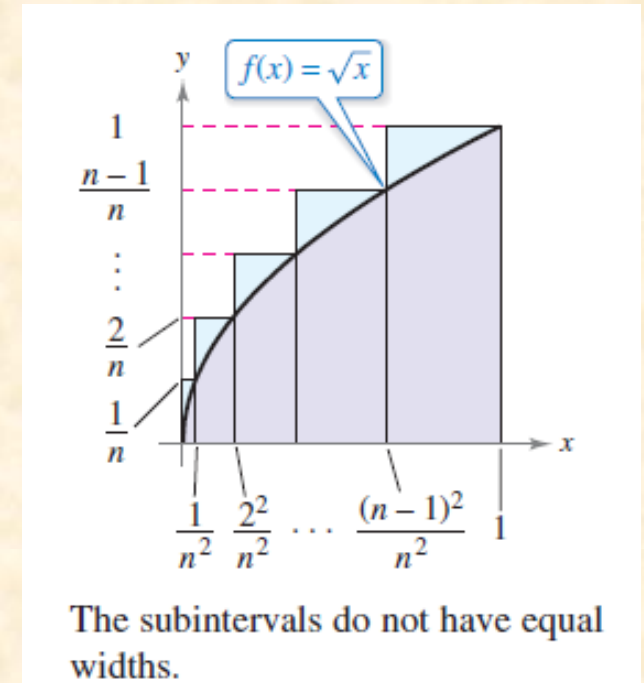


Figure 4.18

Example 1 – Solution

The width of the i th interval is

$$\Delta x_i = \frac{i^2}{n^2} - \frac{(i-1)^2}{n^2}$$

$$= \frac{i^2 - i^2 + 2i - 1}{n^2}$$

$$= \frac{2i - 1}{n^2}.$$

So, the limit is

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{\frac{i^2}{n^2}} \left(\frac{2i - 1}{n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{i=1}^n (2i^2 - i)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^3} \left[2 \left(\frac{n(n+1)(2n+1)}{6} \right) - \frac{n(n+1)}{2} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{4n^3 + 3n^2 - n}{6n^3}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{2}{3} + \frac{1}{2n} - \frac{1}{6n^2} \right)$$

$$= \frac{2}{3}.$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

Riemann Sums

This agrees with the limit found in Example 1, even though that example used a partition having subintervals of unequal widths.

The reason this particular partition gave the proper area is that as n increases, the *width of the largest subinterval approaches zero*.

This is a key feature of the development of definite integrals.

Riemann Sums

In the definition of a Riemann sum below, note that the function f has no restrictions other than being defined on the interval $[a, b]$.

Definition of Riemann Sum

Let f be defined on the closed interval $[a, b]$, and let Δ be a partition of $[a, b]$ given by

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

where Δx_i is the width of the i th subinterval

$$[x_{i-1}, x_i]. \quad \text{\textit{ith subinterval}}$$

If c_i is *any* point in the i th subinterval, then the sum

$$\sum_{i=1}^n f(c_i) \Delta x_i, \quad x_{i-1} \leq c_i \leq x_i$$

is called a **Riemann sum** of f for the partition Δ .

Riemann Sums

The width of the largest subinterval of a partition Δ is the **norm** of the partition and is denoted by $\|\Delta\|$. If every subinterval is of equal width, the partition is **regular** and the norm is denoted by

$$\|\Delta\| = \Delta x = \frac{b - a}{n}.$$

Regular partition

For a **general partition**, the norm is related to the number of subintervals of $[a, b]$ in the following way.

$$\frac{b - a}{\|\Delta\|} \leq n$$

General partition

Riemann Sums

As shown in Figure 4.20, for any positive value of n , the norm of the partition Δ_n is $\frac{1}{2}$.

So, letting n approach infinity does not force $\|\Delta\|$ to approach 0. In a regular partition, however, the statements $\|\Delta\| \rightarrow 0$ and $n \rightarrow \infty$ are equivalent.

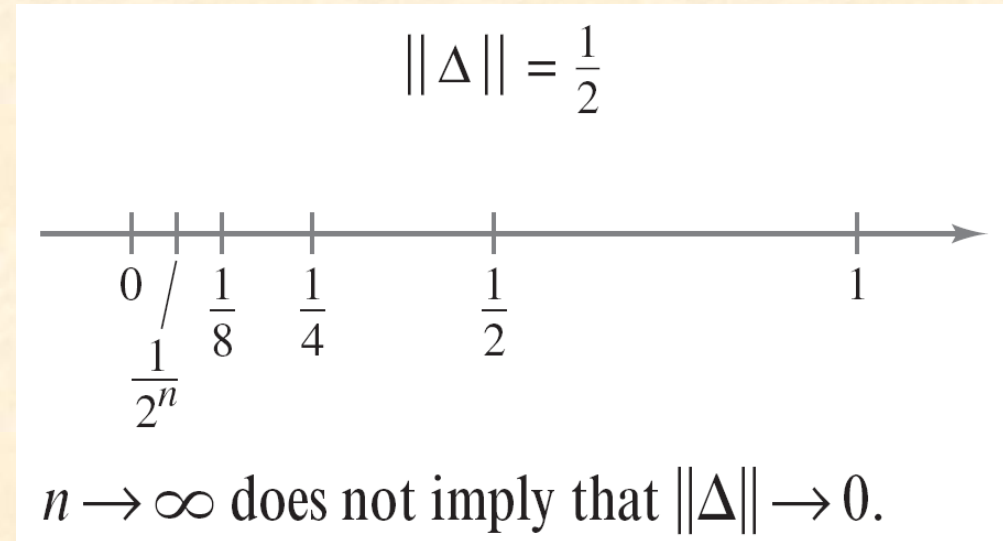


Figure 4.20

Definite Integrals

Definition of Definite Integral

If f is defined on the closed interval $[a, b]$ and the limit of Riemann sums over partitions Δ

$$\lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n f(c_i) \Delta x_i$$

exists (as described above), then f is said to be **integrable** on $[a, b]$ and the limit is denoted by

$$\lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n f(c_i) \Delta x_i = \int_a^b f(x) dx.$$

The limit is called the **definite integral** of f from a to b . The number a is the **lower limit** of integration, and the number b is the **upper limit** of integration.

Example 2 – Evaluating a Definite Integral as a Limit

Evaluate the definite integral

$$\int_{-2}^1 2x \, dx.$$

Solution:

The function $f(x) = 2x$ is integrable on the interval $[-2, 1]$ because it is continuous on $[-2, 1]$.

Moreover, the definition of integrability implies that any partition whose norm approaches 0 can be used to determine the limit.

Example 2 – Solution

For computational convenience, define Δ by subdividing $[-2, 1]$ into n subintervals of equal width

$$\Delta x_i = \Delta x = \frac{b - a}{n} = \frac{3}{n}.$$

Choosing c_i as the right endpoint of each subinterval produces

$$c_i = a + i(\Delta x) = -2 + \frac{3i}{n}.$$

Example 2 – Solution

So, the definite integral is

$$\int_{-2}^1 2x \, dx = \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n f(c_i) \Delta x_i$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n 2 \left(-2 + \frac{3i}{n} \right) \left(\frac{3}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{6}{n} \sum_{i=1}^n \left(-2 + \frac{3i}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{6}{n} \left(-2 \sum_{i=1}^n 1 + \frac{3}{n} \sum_{i=1}^n i \right)$$

$$= \lim_{n \rightarrow \infty} \frac{6}{n} \left\{ -2n + \frac{3}{n} \left[\frac{n(n+1)}{2} \right] \right\}$$

$$= \lim_{n \rightarrow \infty} \left(-12 + 9 + \frac{9}{n} \right)$$

$$= -3.$$

Example 2 – Solution

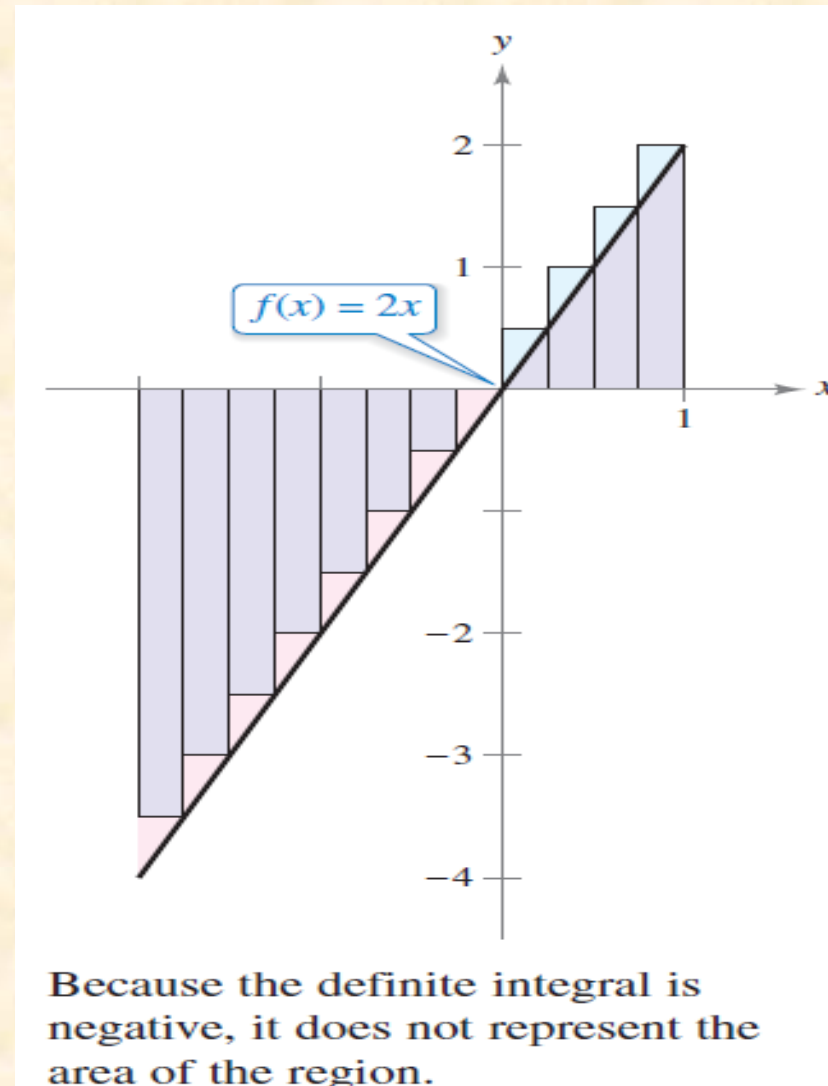


Figure 4.21

Properties of Definite Integrals

Definitions of Two Special Definite Integrals

1. If f is defined at $x = a$, then $\int_a^a f(x) \, dx = 0$.
2. If f is integrable on $[a, b]$, then $\int_b^a f(x) \, dx = -\int_a^b f(x) \, dx$.

THEOREM 4.6 Additive Interval Property

If f is integrable on the three closed intervals determined by a , b , and c , then

$$\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx. \quad \text{See Figure 4.25.}$$

Properties of Definite Integrals

THEOREM 4.7 Properties of Definite Integrals

If f and g are integrable on $[a, b]$ and k is a constant, then the functions kf and $f \pm g$ are integrable on $[a, b]$, and

$$1. \int_a^b kf(x) \, dx = k \int_a^b f(x) \, dx$$

$$2. \int_a^b [f(x) \pm g(x)] \, dx = \int_a^b f(x) \, dx \pm \int_a^b g(x) \, dx.$$

Properties of Definite Integrals

THEOREM 4.8 Preservation of Inequality

1. If f is integrable and nonnegative on the closed interval $[a, b]$, then

$$0 \leq \int_a^b f(x) dx.$$

2. If f and g are integrable on the closed interval $[a, b]$ and $f(x) \leq g(x)$ for every x in $[a, b]$, then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

The Fundamental Theorem of Calculus

THEOREM 4.9 The Fundamental Theorem of Calculus

If a function f is continuous on the closed interval $[a, b]$ and F is an antiderivative of f on the interval $[a, b]$, then

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

The Mean Value Theorem for Integrals

THEOREM 4.10 Mean Value Theorem for Integrals

If f is continuous on the closed interval $[a, b]$, then there exists a number c in the closed interval $[a, b]$ such that

$$\int_a^b f(x) \, dx = f(c)(b - a).$$

Average Value of a Function

Definition of the Average Value of a Function on an Interval

If f is integrable on the closed interval $[a, b]$, then the average value of f on the interval is

$$\frac{1}{b - a} \int_a^b f(x) \, dx.$$

See Figure 4.32.

Example 4 – Finding the Average Value of a Function

Find the average value of $f(x) = 3x^2 - 2x$ on the interval $[1, 4]$.

Solution:

The average value is

$$\frac{1}{b-a} \int_a^b f(x) dx = \frac{1}{4-1} \int_1^4 (3x^2 - 2x) dx$$

$$= \frac{1}{3} \left[x^3 - x^2 \right]_1^4$$

$$= \frac{1}{3} [64 - 16 - (1 - 1)]$$

$$= \frac{48}{3} = 16.$$

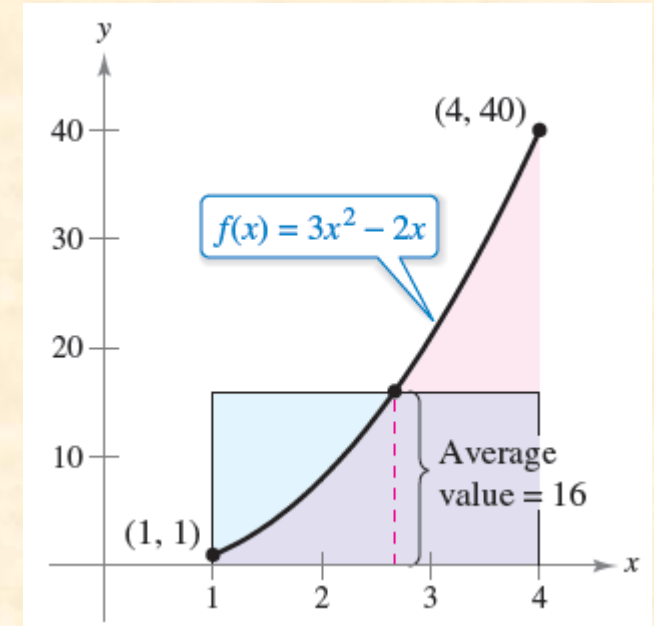
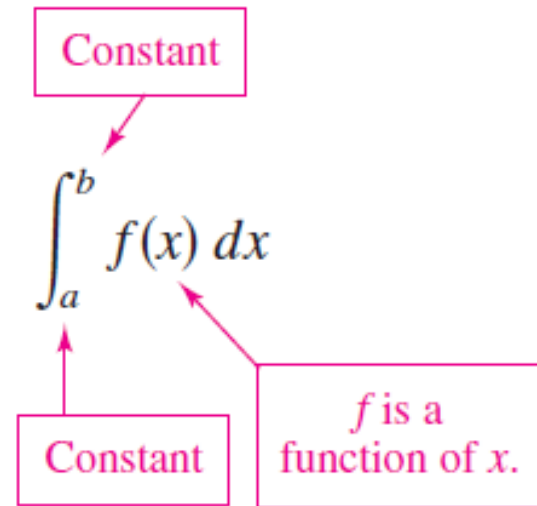


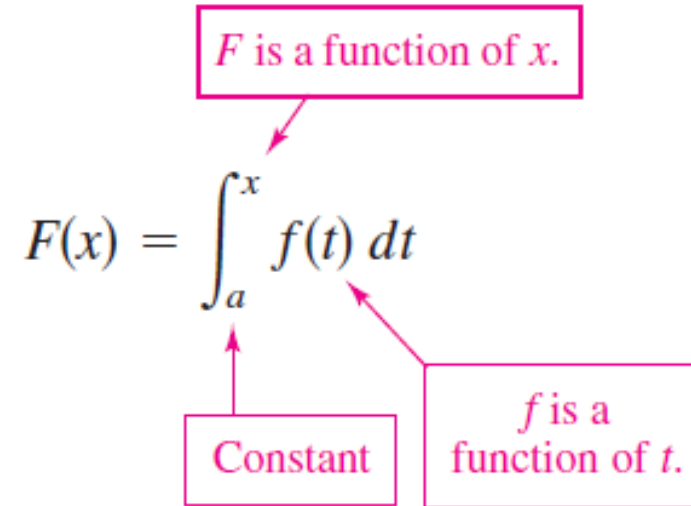
Figure 4.33

The Second Fundamental Theorem of Calculus

The Definite Integral as a Number



The Definite Integral as a Function of x



Example 6 – The Definite Integral as a Function

Evaluate the function

$$F(x) = \int_0^x \cos t \, dt$$

at $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3},$ and $\frac{\pi}{2}.$

Solution:

You could evaluate five different definite integrals, one for each of the given upper limits.

However, it is much simpler to fix x (as a constant) temporarily to obtain

$$\int_0^x \cos t \, dt = \sin t \Big|_0^x = \sin x - \sin 0 = \sin x.$$

The Second Fundamental Theorem of Calculus

THEOREM 4.11 The Second Fundamental Theorem of Calculus

If f is continuous on an open interval I containing a , then, for every x in the interval,

$$\frac{d}{dx} \left[\int_a^x f(t) \, dt \right] = f(x).$$

Example 7 – The Second Fundamental Theorem of Calculus

Evaluate $\frac{d}{dx} \left[\int_0^x \sqrt{t^2 + 1} \, dt \right].$

Solution:

Note that $f(t) = \sqrt{t^2 + 1}$ is continuous on the entire real number line.

So, using the Second Fundamental Theorem of Calculus, you can write

$$\frac{d}{dx} \left[\int_0^x \sqrt{t^2 + 1} \, dt \right] = \sqrt{x^2 + 1}.$$

Thanks a lot ...